

Exam structure:

The test will be 3 hours, on paper, in person on campus.

This test will be sat by students who have already received a credit, i.e. 65% of the unit. The objective of this test is to distinguish between students who should receive:

- A credit - they should score 0-4 on this test
- A distinction - they should score 5-14 on this test
- A high distinction - they should score 15-35 on this test

As you can see, more than half of the marks on the test should only be obtained by students who are at a high distinction level. As a consequence, you can expect the easiest questions on the test to be challenging, and the hardest questions to be very hard. It is likely that some students who attempt the test will receive very few marks.

Another way to say this is that you "only" need 43% on this test to get an HD. So you should not expect to feel the same way walking out of this test as a typical exam, i.e. that you only did well if you answered most questions.

With this in mind, it is important to remember that you have done very well to be sitting this test at all! This is a difficult unit, and if you are sitting this test, you have already passed and received a credit. Given that you have made it this far, you might as well give it your best shot and see how you go.

Test structure

Section 1 - 10 marks

- 10 questions each worth 1 mark.
- Assess your ability to answer simple definition/justification type questions about the content.
- Answers will be short, 1-3 sentences (sometimes just a number or formula)
- **Inaccurate** or **vague** answers will receive 0, **clear and correct** answers will receive 1.

Section 2 - 10 marks

- 5 questions each worth 2 marks.
- Assess your ability to apply your understanding of the content.
- Proofs or problems solving questions
- Clearly related to a concept/topic from the unit
- Answers will be 3-5 sentences
- Partial marks (i.e. 1 mark out of 2) are possible for answers with the right idea but some issue

Section 3 - 15 marks

- 3 questions each worth 5 marks
- More than 3 questions will be provided
- You must choose **which three** you answer
- If you answer more than 3, we will mark 3 at random. We will not look at the fourth submission at all
- Assess your ability to solve novel problems **using concepts/algorithms taught in this unit**.
- It may not be initially clear how to solve these problems!
- Partial marks are possible

Time

It is vital to spend a few minutes **reading the whole test** before starting to work. You probably should not work through the test from the front to the back, but rather in order of questions that you are confident you can answer well.

Remember that you only need 5 marks for a distinction! You can find those 5 marks anywhere in the test, they might not be on the first 5 questions.

The test runs for 3 hours and has 35 marks. That said, we do not expect all the marks to take the same amount of time. How long it takes to do 1 mark worth of work depends on your skillset, but in general the earlier marks will take less time.

Logistic

To be announced closer to the date, once we have the full list of students who will be sitting for it after the Week-14 PTs.

Some **very important** notes on the list of practice questions

1. Questions are listed with a section. This does **not** mean the question is of the correct **difficulty** for that section, some may be far easier. This does **not** mean the question is of the **type** that would be found in that section. What it means is that it requires skills or knowledge that would be tested in that section.
2. We have included most questions that we thought were relevant to the D/HD test. The questions in the applied sheets do not cover every topic equally, so there may be more questions than you might need on some topics. You should not proceed through the list from front to back, but in an order that makes sense to you based on your skills.
3. Many questions on the applied sheets are more open ended than questions on the test will be, as they are designed for a discussion based context. In order to help you prepare for the test, we recommend that you
 1. Determine the complexity of every algorithm you create while answering questions
 2. Justify/prove every claim that you make to yourself while answering questions

The questions in bold are a suggestion for questions that cover key ideas very well.

Week 1

- Problem 2 - section 2
- Problem 3 - section 2
- problem 5 - section 2
- Problem 6 - section 1
- Problem 8 - section 1

Week 2

- problem 2 - section 1
- problem 3 - section 2
- problem 4 - section 1
- problem 5 - section 2 or 3
- problem 6 - section 3
- problem 8 - section 1
- problem 9 - section 1
- problem 13 - section 2
- problem 16a - section 1
- problem 16b - section 1
- problem 16d - section 1

Week 3

- problem 1 - section 2
- problem 2 - section 2
- **problem 3 - section 2**
- problem 4 - section 2 or 3
- problem 5 - section 2
- problem 6 - section 2

- problem 7 - section 3
- problem 8 - section 1
- problem 9 - section 2
- problem 10 - section 2
- problem 11 - section 1

Week 4

- problem 2 - section 2
- problem 3 - section 2
- problem 4 - section 2
- problem 5 - section 2
- problem 5 - section 2
- problem 6 - section 2
- problem 7 - section 2
- problem 9.2 - section 3

Week 5

- problem 3a - section 1
- problem 3b - section 2
- **problem 5 - section 2**
- problem 6 - section 2
- problem 7 - section 2
- problem 10 - section 1
- problem 11 - section 3
- problem 12 - section 2
- problem 13 - section 3

Week 6

- problem 1 - section 1
- problem 2 - section 1
- problem 3 - section 3
- problem 4 - section 3
- problem 5a - section 1
- problem 5b - section 2
- problem 6 - section 2
- problem 7 - section 2
- problem 8 - section 2
- problem 9 - section 2
- **problem 10 - section 2**

Week 7

- problem 1 - section 1
- problem 2 - section 1
- problem 3 - section 2
- problem 4 - section 2
- problem 5 - section 2
- problem 6 - section 2
- problem 7 - section 2
- problem 8 - section 3
- problem 9a - section 1
- problem 9b - section 3
- There are probably more good practice question in here too

Week 8

- problem 1 - section 1
- problem 3 - section 2
- problem 4 - section 1

- problem 5 - section 2
- problem 6 - section 2
- problem 7c - section 2
- problem 9 - section 2
- problem 10 - section 3
- **problem 11b - section 1**
- problem 11c - section 2

Week 9

- problem 1 - section 1
- problem 2 - section 1
- problem 3 - section 1
- problem 4 - section 2
- problem 5 - section 2
- problem 6 - section 2
- problem 8 - section 1
- problem 9 - section 2
- problem 10 - section 3
- **problem 11 - section 3**

Week 10

- problem 2 - section 2
- problem 3 - section 2
- problem 7 - section 2

Week 11

- problem 1 - section 1
- problem 3 - section 2
- problem 4 - section 3
- problem 5 - section 2
- section 6 - section 3
- problem 7 - section 3

Week 12

- problem 2 - section 3
- problem 3 - section 2
- problem 4 - section 2
- problem 5 - section 3
- section 6 - section 3